

Autonomous and Non-Autonomous AI Systems

Part 1: Semi-supervised learning with HITL (Human-in-the-Loop) Systems

by Steven Miers

◆ Component Guide					
Mode	Op	2a	Out	Role	
● Product	Auto	Model-pkg	Predictions	ML Eng	
■ User	User?	Predictions	Results	DB Eng	
■ Data	Auto	Results	Feedback	DB Eng	
○ Curator	Human	Feedback	Curated	Curator	
● Pipeline	Auto	Curated	Test data	DB Eng	
● Training	Auto	Test data	APRINT	ML Eng	
○ Release	Human	APRINT	Approved	Curator	

Legend: ● data source only, ○ human data

► Agentic AI in HITL

- Data Collection**
Agents query users, synthesize behavioral signals, generate synthetic data (HITL).
- NOT a Classifier**
Agentic collection does NOT change HITL/HOTL classification, it changes what feeds INTO the Data Curation Service.
- Still Gated**
An agentic collector in HITL still routes to a human curator. The agent is a tool — not the gate.

Node Classification

- **LOOP NODE**
Automated actions, none without human decisions
- **HUMAN GATE**
Human must approve before flow continues
- **DATA / SURFACE**
Stores or surfaces data, handles external requests
- **PIPELINE / TRAINING**
Processes data, performs internal machine learning, distribution
- **BOOTSTRAP**
One-time setup, bypassed after first run

Where's the Human?

Not the End Users...
End users generate signal — they are a data source, not a gate.
► Curator filters training data
► Reviewer approves releases

The Gatekeepers
Decides what happens to data next:
► Curator filters training data
► Reviewer approves releases

\$ the_math.sh
users = 1,000,000
curator_team = 2
the 2-person team is # where the loop lives.

HITL Failure Modes

- ▲ **Throughput Bottleneck**
Humans cap pipeline rate. 90% of ML practitioners cite annotation as boring constraint.
- ▲ **Amplifier Bias**
Systematic degradation by human error. Curator's bias is amplified by poor production agent.
- ▲ **Cost at Scale**
MLAP + Computational AI are huge maintenance alternatives.

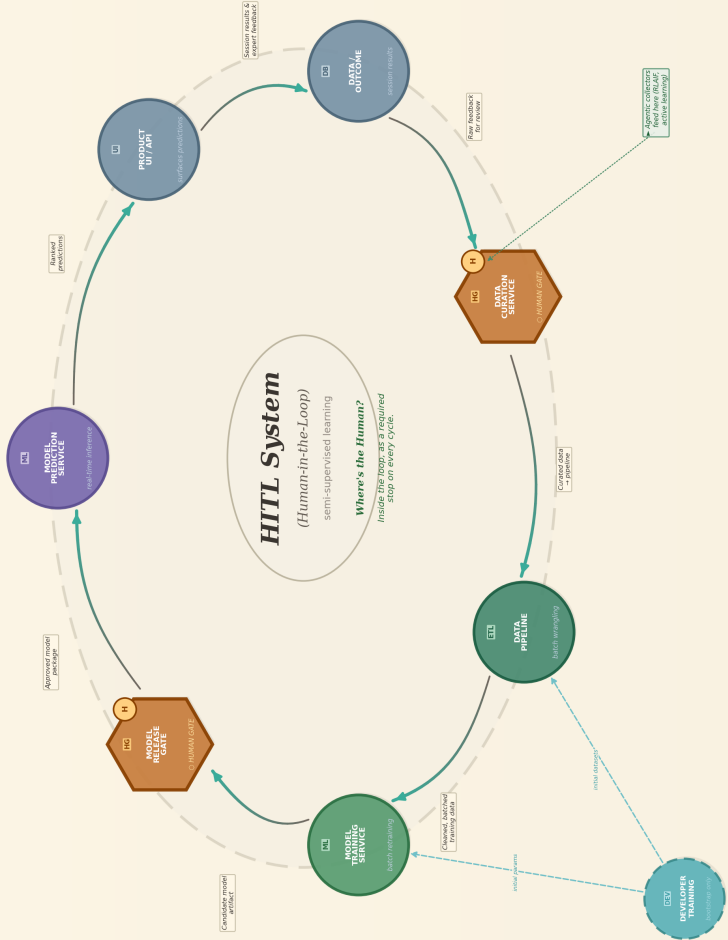
◆ Autonomy Debt

1 Definition
"The gap between what your system did and what you actually want it to do, given human intervention."

2 As Data Scales
Headline: ~ tens of thousands of watchable items. The debt compounds.

3 Three Bottlenecks
The humans who make the system trustworthy also make it slow.
► Curator = thoughtful cap
► Sequence = deploy blocker
► Tendency latencies = stale models

4 Key Insight
The humans who make the system trustworthy also make it slow.



Many production AI systems are loop-shaped, but not closed-loop. Human feedback, curation, and governance introduce delay, filtering, and control boundaries that prevent true autonomy.